



مدرسة جيمس نيو ميلينيوم الخيل
GEMS New Millennium School
AL KHAIL

Netflix Titles Analyser

PROJECT REPORT COMPUTER SCIENCE GRADE XII (2025-26)

Submitted in partial fulfillment of the requirement of CBSE, Delhi

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CERTIFICATE OF COMPLETION

This is to certify that the project titled "Insert Title within Quotes" submitted by Insert Name in partial fulfillment of the requirement for the Central Board of Secondary Education of Grade 12, GEMS New Millennium School is done by him/her during the academic year 2025-26, under my supervision and guidance and this project or any part thereof has not been submitted elsewhere for any classes.

Signature of Internal Guide

Ms.

Date:

Submitted for the project evaluation held on _____ 2026 and viva voce held on _____, at GEMS New Millennium School.

Signature of the External Examiner

ACKNOWLEDGEMENT

*I take this opportunity to sincerely express
my gratitude to all who have helped me to
complete this
project file.*

*My sincere thanks to
Mrs. Fatima Martin,
our Principal
for her constant support and encouragement
and*

*Ms. Edith Paulin,
our Grade XII Computer Science teacher,
whose help, guidance and advice
enabled me to complete this project file.*



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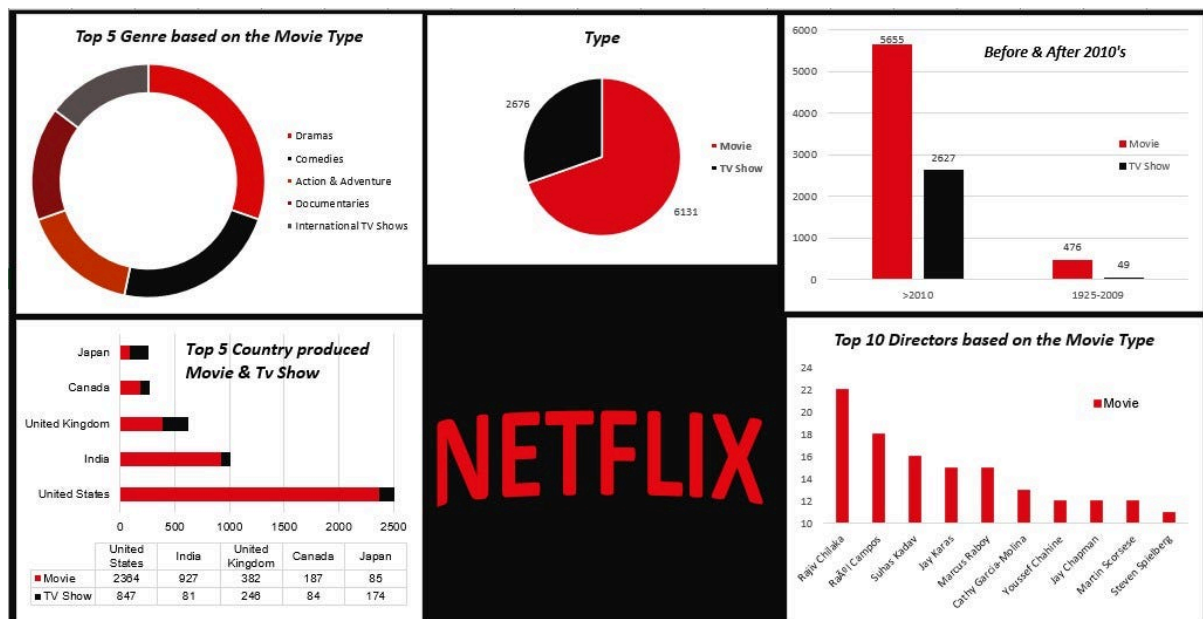
BACKGROUND OF THE PROJECT

In today's world of digital streaming, platforms like Netflix host thousands of movies and TV shows. However, browsing through such a large catalog is often challenging.

This project, **Netflix Titles Analyser**, allows users to:

- 1) Search and filter titles based on genre, country, rating, release year, or keywords.
- 2) Add new titles and delete existing ones.
- 3) View visualisations such as top countries, rating distribution, and movies vs. TV shows.

The aim is to demonstrate Database Management (SQL) integrated with Python to create a functional application.



FUNCTIONS AND MODULES

Modules Used

- `sqlite3`: To connect Python with SQLite database.
- `pandas`: To read and clean Netflix CSV dataset.
- `matplotlib`: For generating visualisations.
- `tabulate`: For neat tabular display of search results.

Functions

- `advanced_search()`: Filters records by one or multiple criteria (genre, country, rating, year, keyword).
- `add_record()`: Adds a new Netflix title to the database.
- `delete_record()`: Deletes a record by show ID.
- `show_visualisations()`: Displays bar charts, pie charts, and line graphs for insights.
- `search_by_genre()`: Filters records by their genre.
- `search_by_country()`: Filters records by their country.
- `search_by_rating()`: Filters records by their rating.
- `search_by_year()`: Filters records by their year.
- `search_by_title()`: Filters records by their title or keywords

FLOW OF THE PROJECT

1. Download the Netflix titles dataset from Kegggle.
2. Load the dataset (`netflix_titles.csv`) into SQLite (`netflix.db`).
3. Create a menu-driven program with the following options:
 - Search by Genre
 - Search by Country
 - Search by Rating
 - Search by Release Year
 - Search by Title Keyword
 - Advanced Search (multiple filters)
 - Add Record
 - Delete Record
 - Show Visualisations
 - Exit
4. Fetch and display results in tables.
5. Generate visual insights using Matplotlib.

USE OF TECHNOLOGY

Python

Python is a high-level, interpreted programming language that emphasizes readability and simplicity. It is widely used for database applications, data analysis, and visualisations.

SQLite

SQLite is a lightweight relational database built into Python. It stores data in a single file ([netflix.db](#)) and supports standard SQL commands.

Matplotlib

Matplotlib is a plotting library in Python for creating static, animated, and interactive visualisations. It is used here to show data distributions (ratings, countries, etc.).

Pandas

Pandas is a Python library for handling tabular data. It is used to load and preprocess the Netflix dataset.

Hardware & Software used:

- **System:** Macbook Air M3
- **OS:** macOS
- **Software:** Python 3.11, SQLite, Visual Studio Code
- **Libraries:** sqlite3, pandas, matplotlib, tabulate

SOURCE CODE

```
import sqlite3
from tabulate import tabulate
import matplotlib.pyplot as plt

def advanced_search(genre=None, country=None, rating=None, year=None,
keyword=None):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()

    query = "SELECT title, release_year, rating, country, listed_in
FROM netflix WHERE 1=1"
    params = []

    if genre:
        query += " AND listed_in LIKE ?"
        params.append('%' + genre + '%')
    if country:
        query += " AND country LIKE ?"
        params.append('%' + country + '%')
    if rating:
        query += " AND rating = ?"
        params.append(rating)
    if year:
        query += " AND release_year = ?"
        params.append(year)
    if keyword:
        query += " AND title LIKE ?"
        params.append('%' + keyword + '%')

    cursor.execute(query, tuple(params))
    results = cursor.fetchall()
    conn.close()
    return results

def show_visualisations():
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()

    cursor.execute("SELECT country, COUNT(*) FROM netflix WHERE country
NOT NULL GROUP BY country ORDER BY COUNT(*) DESC LIMIT 10")
    data = cursor.fetchall()
    if data:
        countries, counts = zip(*data)
        plt.figure(figsize=(8,5))
        plt.bar(countries, counts)
        plt.title("Top 10 Countries with Most Titles")
```

```

        plt.xticks(rotation=45)
        plt.show()

    cursor.execute("SELECT rating, COUNT(*) FROM netflix GROUP BY
rating ORDER BY COUNT(*) DESC LIMIT 10")
    data = cursor.fetchall()
    if data:
        ratings, counts = zip(*data)
        plt.figure(figsize=(8,5))
        plt.bar(ratings, counts, color="orange")
        plt.title("Distribution of Ratings")
        plt.xticks(rotation=45)
        plt.show()

    cursor.execute("SELECT type, COUNT(*) FROM netflix GROUP BY type")
    data = cursor.fetchall()
    if data:
        labels, counts = zip(*data)
        plt.figure(figsize=(5,5))
        plt.pie(counts, labels=labels, autopct='%1.1f%%')
        plt.title("Movies vs TV Shows")
        plt.show()

    cursor.execute("SELECT release_year, COUNT(*) FROM netflix GROUP BY
release_year ORDER BY release_year")
    data = cursor.fetchall()
    if data:
        years, counts = zip(*data)
        plt.figure(figsize=(10,5))
        plt.plot(years, counts, marker='o')
        plt.title("Number of Titles by Year")
        plt.xlabel("Year")
        plt.ylabel("Count")
        plt.show()

    conn.close()

def search_by_genre(genre):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()
    cursor.execute("SELECT title, release_year, rating FROM netflix
WHERE listed_in LIKE ?", ('%' + genre + '%',))
    results = cursor.fetchall()
    conn.close()
    return results

def search_by_country(country):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()
    cursor.execute("SELECT title, release_year, rating FROM netflix
WHERE country LIKE ?", ('%' + country + '%',))
    results = cursor.fetchall()
    conn.close()
    return results

```

```

def search_by_rating(rating):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()
    cursor.execute("SELECT title, release_year FROM netflix WHERE
rating = ?", (rating,))
    results = cursor.fetchall()
    conn.close()
    return results

def search_by_year(year):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()
    cursor.execute("SELECT title, rating FROM netflix WHERE
release_year = ?", (year,))
    results = cursor.fetchall()
    conn.close()
    return results

def search_by_title(keyword):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()
    cursor.execute("SELECT title, release_year, rating FROM netflix
WHERE title LIKE ?", ('%' + keyword + '%',))
    results = cursor.fetchall()
    conn.close()
    return results

def add_record(record):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()
    cursor.execute("""
INSERT INTO netflix (show_id, type, title, director, cast, country,
date_added, release_year, rating, duration, listed_in, description)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
""", record)
    conn.commit()
    conn.close()
    print("Record added successfully!")

def delete_record(show_id):
    conn = sqlite3.connect("netflix.db")
    cursor = conn.cursor()
    cursor.execute("DELETE FROM netflix WHERE show_id = ?", (show_id,))
    conn.commit()
    conn.close()
    print("Record deleted successfully!")

while True:
    print("\nNetflix Titles Analyser")
    print("1. Search by Genre")
    print("2. Search by Country")
    print("3. Search by Rating")
    print("4. Search by Release Year")
    print("5. Search by Title Keyword")

```

```

print("6. Add Record")
print("7. Delete Record")
print("8. Advanced Search")
print("9. Show Visualisations")
print("10. Exit")

choice = input("Enter choice: ")

if choice == "1":
    genre = input("Enter genre: ")
    results = search_by_genre(genre)
    print(tabulate(results[:10], headers=["Title", "Year",
"Rating"], tablefmt="grid"))

    elif choice == "2":
        country = input("Enter country: ")
        results = search_by_country(country)
        print(tabulate(results[:10], headers=["Title", "Year",
"Rating"], tablefmt="grid"))

    elif choice == "3":
        rating = input("Enter rating (e.g., PG, TV-MA): ")
        results = search_by_rating(rating)
        print(tabulate(results[:10], headers=["Title", "Year",
"Rating"], tablefmt="grid"))

    elif choice == "4":
        year = int(input("Enter release year: "))
        results = search_by_year(year)
        print(tabulate(results[:10], headers=["Title", "Year",
"Rating"], tablefmt="grid"))

    elif choice == "5":
        keyword = input("Enter part of title: ")
        results = search_by_title(keyword)
        print(tabulate(results[:10], headers=["Title", "Year",
"Rating"], tablefmt="grid"))

    elif choice == "6":
        print("Enter new record details:")
        record = (
            input("show_id: "),
            input("type: "),
            input("title: "),
            input("director: "),
            input("cast: "),
            input("country: "),
            input("date_added: "),
            int(input("release_year: ")),
            input("rating: "),
            input("duration: "),
            input("listed_in: "),
            input("description: ")
        )
        add_record(record)

```

```

elif choice == "7":
    show_id = input("Enter show_id to delete: ")
    delete_record(show_id)

elif choice == "8":
    print("\nEnter filters (leave blank to skip):")
    genre = input("Genre: ").strip()
    country = input("Country: ").strip()
    rating = input("Rating (e.g., PG, TV-MA): ").strip()
    year = input("Release Year: ").strip()
    keyword = input("Keyword in Title: ").strip()

    year = int(year) if year else None

    results = advanced_search(
        genre if genre else None,
        country if country else None,
        rating if rating else None,
        year,
        keyword if keyword else None
    )

    if results:
        print(tabulate(results[:15], headers=["Title", "Year",
"Rating", "Country", "Genre"], tablefmt="grid"))
    else:
        print("No results found.")

elif choice == "9":
    show_visualisations()

elif choice == "10":
    print("Exiting Netflix Analyser. Goodbye!")
    break

else:
    print("Invalid choice. Try again.")

```

SAMPLE OUTPUT

```
Netflix Titles Analyser
1. Search by Genre
2. Search by Country
3. Search by Rating
4. Search by Release Year
5. Search by Title Keyword
6. Add Record
7. Delete Record
8. Advanced Search
9. Show Visualisations
10. Exit
Enter choice: 1
Enter genre: Horror
```

Title	Year	Rating
Midnight Mass	2021	TV-MA
Dark Skies	2013	PG-13
Jaws 2	1978	PG
Jaws 3	1983	PG
Jaws: The Revenge	1987	PG-13
Krishna Cottage	2004	TV-14
Ragini MMS	2011	TV-MA
Ragini MMS 2	2014	TV-MA
The Old Ways	2020	TV-MA
Boomika	2021	TV-14

```
Netflix Titles Analyser
1. Search by Genre
2. Search by Country
3. Search by Rating
4. Search by Release Year
5. Search by Title Keyword
6. Add Record
7. Delete Record
8. Advanced Search
9. Show Visualisations
10. Exit
Enter choice: 8
```

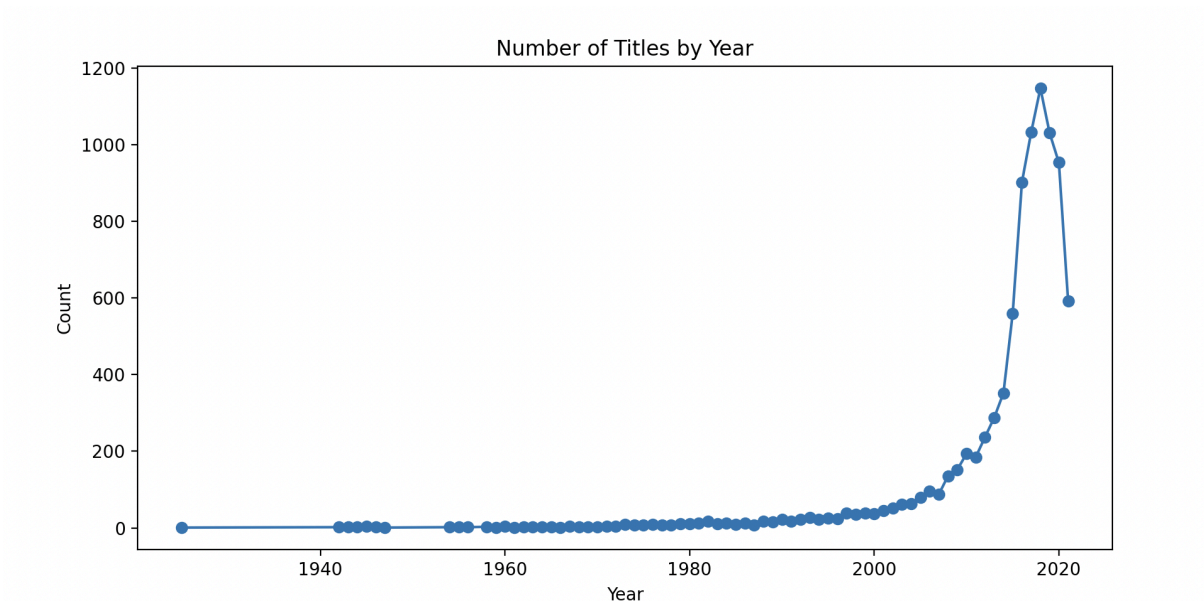
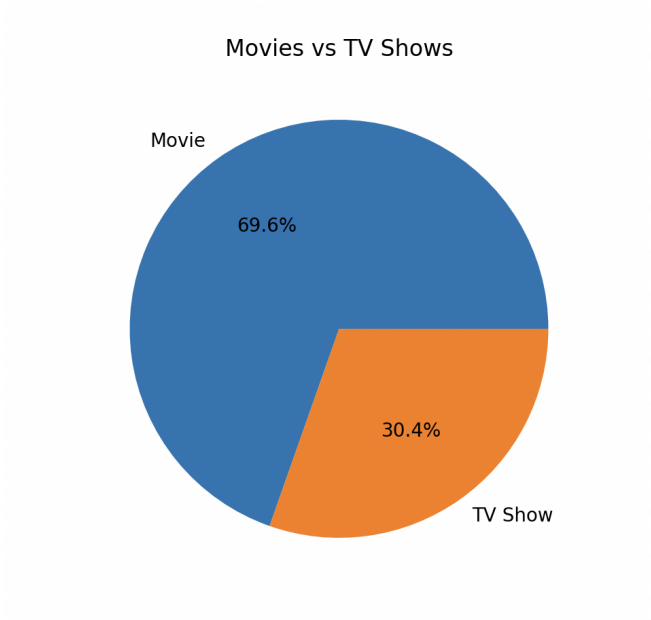
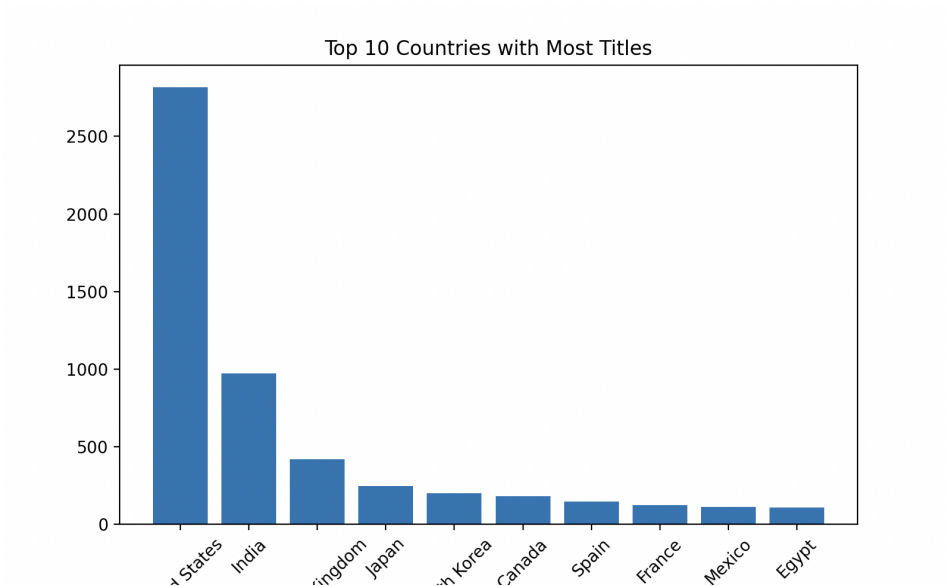
```
Enter filters (leave blank to skip):
Genre: comedy
Country: India
Rating (e.g., PG, TV-MA):
Release Year: 2020
Keyword in Title:
```

Title	Year	Rating	Country	Genre
Vir Das: Outside In - The Lockdown Special	2020	TV-MA	India	Stand-Up Comedy
Kenny Sebastian: The Most Interesting Person in the Room	2020	TV-MA	India	Stand-Up Comedy
Yours Sincerely, Kanan Gill	2020	TV-MA	India	Stand-Up Comedy
Vir Das: For India	2020	TV-MA	India	Stand-Up Comedy

```
Netflix Titles Analyser
1. Search by Genre
2. Search by Country
3. Search by Rating
4. Search by Release Year
5. Search by Title Keyword
6. Add Record
7. Delete Record
8. Advanced Search
9. Show Visualisations
10. Exit
Enter choice: 6
Enter new record details:
show_id: s9998
type: Movie
title: Inception
director: Christopher Nolan
cast: Leonardo DiCaprio, Joseph Gordon-Levitt, Ellen Page
country: United States
date_added: April 1, 2021
release_year: 2010
rating: PG-13
duration: 148 min
listed_in: Sci-Fi & Fantasy, Action
description:
Record added successfully!
```

```
Netflix Titles Analyser
1. Search by Genre
2. Search by Country
3. Search by Rating
4. Search by Release Year
5. Search by Title Keyword
6. Add Record
7. Delete Record
8. Advanced Search
9. Show Visualisations
10. Exit
Enter choice: 7
Enter show_id to delete: Jaws
Record deleted successfully!
```

```
Netflix Titles Analyser
1. Search by Genre
2. Search by Country
3. Search by Rating
4. Search by Release Year
5. Search by Title Keyword
6. Add Record
7. Delete Record
8. Advanced Search
9. Show Visualisations
10. Exit
Enter choice: 9
█
```



CONCLUSION

The project Netflix Titles Analyser successfully demonstrates how Python and SQL can be integrated to manage and analyse large datasets.

Through this project, I learned:

- Basics of database design (tables, queries, CRUD operations).
- Connecting Python with SQL (sqlite3 module).
- Data cleaning and insertion using Pandas.
- Displaying results in a user-friendly format.
- Using Matplotlib to generate visual insights.

The project highlights the importance of database-driven applications in the real world and shows how Python makes it easy to work with data effectively.

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